Carolina Hoyos Agudo

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Education

Universidad Complutense de Madrid — **Madrid**
PhD in Computer Engineering — *Oct 2026 – Present*

Universidad Complutense de Madrid — **Madrid**
MSc in Formal Methods in Computer Engineering — *Oct 2025 – Sep 2026*
Highest honors in **Quantum Computing**, **Assisted Program Verification**, **Static Analysis and Constraint Solving**, **Bioinspired Algorithm Design** and **Theory of Programming Languages**.

Universidad Complutense de Madrid — **Madrid**
BSc in Mathematics — *Jan 2020 – Feb 2025*
Specialization in Computer Science.

Sapienza Università di Roma — **Rome, Italy**
Erasmus Exchange — *2022 – 2023*

IES Arquitecto Ventura Rodríguez — **Boadilla del Monte, Madrid**
Excellence Baccalaureate — *2018 – 2020*

Professional Experience

Universidad Complutense de Madrid — **Madrid**
Research Staff (PLI) — *Jun 2026 – Present*
Contracted researcher on the project *"Rigorous methods for reliable and efficient software systems"*.

Department of Statistics and Operations Research, UCM — **Madrid**
Substitute Lecturer — *Feb 2026 – Present*

- **Statistics** – BSc in Mathematics: 4 ECTS (40 h).
- **Time Series** – Double Degree in Economics–Mathematics and Data Science: 3 ECTS (30 h).
- **Statistical Software** – MSc in Statistical-Computational Information Processing: 0.75 ECTS (7.5 h).
- **Statistics** – BSc in Pharmacy: 1.5 ECTS (15 h).
- **Time Series, Panel Data and Calendar Seasonal Adjustment** – MSc in Official Statistics and Socio-Economic Indicators: 1.15 ECTS (11.5 h).

FADOSS Research Group, UCM — **Madrid**
Curricular Internship — *Feb 2026 – May 2026*
Collaborated on research tasks within the FADOSS group.

Grant Thornton — **Madrid**
IT Audit — *Sep 2024 – Oct 2025*
Took part in information systems audits.

Conferences and Activities

2022 – 2026: Participation in the Spanish National Meeting of Mathematics Students (ENEM).

2024: Participation in EACA (Meeting on Computer Algebra and Applications).

2026: *Marktoberdorf International Summer School* (Herrsching am Ammersee, Germany); *StandUp Math 2026* (UCM); *Emerging Trends in Game Theory* – VII SEIO Course on Game Theory; *SEIO 2026* (Santiago de Compostela).

Technical Skills

Programming: Python, R, Dafny, Lean, Haskell

Tools: LaTeX, Git

Other

Languages: Spanish (native), English (C1), Italian (B1), Chinese (A1.1)